MILITARY STANDARD

MIL-STD-1320-233A (NAVY)

22 APRIL 1993

SUPERSEDING
MIL-STD-1320-233
12 DECEMBER 1988

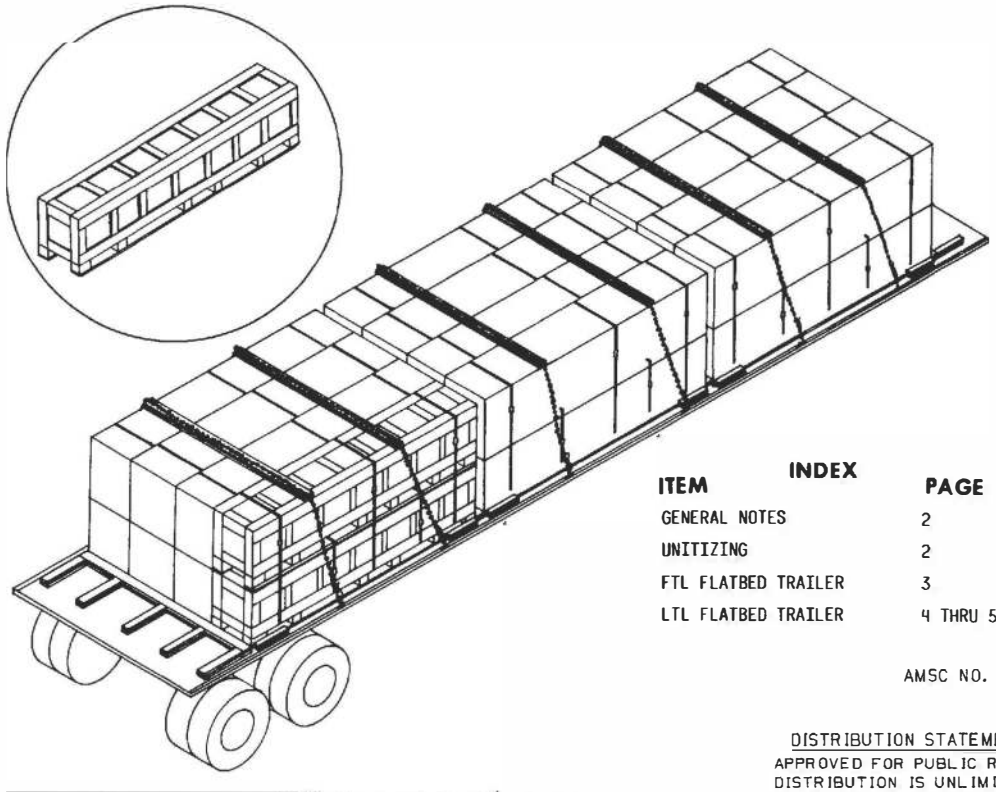
TRUCKLOADING

ROCKET MOTOR FOR SM-2 BLK II MISSILE MK 104, ROCKET MOTOR MK 56 MOD 0, 1, 2, 3, AND SUSTAINER MK 30 MOD 2 IN CONTAINER, MK 693 MOD 0

CONTAINER DATA:

DIMENSIONS..... 126 L X 20 W X 25 H
GROSS WEIGHT..... SEE TABLE I
TARE WEIGHT..... 277 LBS
CUBE..... 36.5 CU FT

FOR HAZARD CLASSIFICATION SEE SW020-AC-SAF-010/020/030



INDEX

ITEM	PAGE
GENERAL NOTES	2
UNITIZING	2
FTL FLATBED TRAILER	3
LTL FLATBED TRAILER	4 THRU 5

AMSC NO. - N/A

DISTRIBUTION STATEMENT A
APPROVED FOR PUBLIC RELEASE: DISTRIBUTION IS UNLIMITED.

REV LTR	REVISION DESCRIPTION	DATE	TDA APPROVAL	SYSCOM APPROVAL
A	ADDED THE USE OF ADDITIONAL ITEMS TO SAME CONTAINER	4/22/93	CR	[illegible]

NOTES:
1. UNLESS OTHERWISE SPECIFIED, DIMENSIONS ARE IN INCHES.
2. FOR CROSS REFERENCE TO ASSOCIATED PALLETIZING, CONTAINERLOADING AND CARLOADING MILITARY STANDARDS, REFER TO INDEX TO STANDARDS, MIL-HDBK-236 (NAVY).

FSC 8140

AUTHORIZED AND RELEASED FOR HIGHWAY SHIPMENT ONLY.

SIGNATURE TDA, WPNSTA EARLE — DATE 12/13/88

SIGNATURE BY DIRECTION — DATE 12/15/88

ORIGINATOR [signature] 12-12-88 DATE

WPNSTA EARLE, NEW JERSEY

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:44Z
Check the source to verify that this is the current version before use.

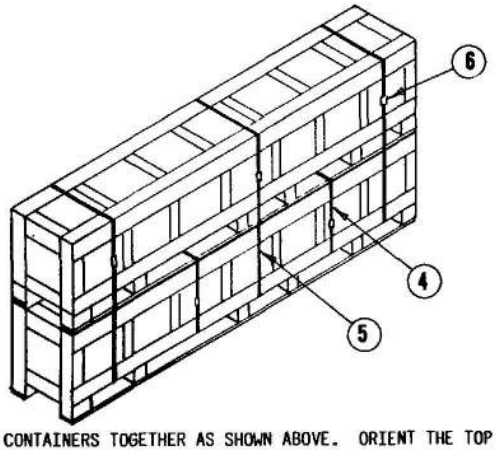
GENERAL NOTES:

1. THIS DOCUMENT GIVES DETAILED INSTRUCTIONS FOR TRUCKLOADING THE DUAL THRUST ROCKET MOTOR (DTRM) MK 104, THE ROCKET MOTOR MK 56 MOD 0, 1, 2, 3, AND THE SUSTAINER MK 30 MOD 2 IN CONTAINER MK 693 MOD 0.
2. THE PROCEDURES DESCRIBED HEREIN ARE INTENDED FOR 40 FT OR LONGER FLATBED TRAILERS. DO NOT USE TRAILERS WITH ALL METAL FLOORS.
3. CHAINS AND STEEL STRAPPING MAY BE USED INTERCHANGEABLY ON A ONE-TO-ONE BASIS FOR TIEDOWNS.
4. STEEL STRAPPING USED AS TIEDOWNS SHALL BE 2 X .050 AND SHALL CONFORM TO AND BE APPLIED AS SPECIFIED IN THE GENERAL TRUCKLOADING DOCUMENT, MIL-STD-1320 (NAVY).
5. CHAINS, FITTINGS, AND LOAD BINDERS SHALL CONFORM TO AND BE APPLIED AS SPECIFIED IN THE GENERAL TRUCKLOADING DOCUMENT, MIL-STD-1320 (NAVY), EXCEPT THAT THE CHAIN/GRAB HOOKS SHALL BE ATTACHED TO THE STAKE POCKETS (NOT AROUND THE RUB RAIL).
6. PROTECTOR BOARDS (DOUBLED 2 X 6'S) SHALL BE USED UNDER THE TIEDOWNS WHEN USING STEEL STRAPPING. THE STRAPS SHALL BE SECURED TO THE PROTECTOR BOARD USING 2-INCH STRAPPING STAPLES. WHEN USING CHAIN, THE CHAINS SHALL BE SECURED TO THE PROTECTOR BOARD BY DRIVING A 10d NAIL THROUGH THE CHAIN LINK AND CLINCHING THE NAIL OVER THE CHAIN. FIVE STAPLES/NAILS SHALL BE USED FOR EACH TIEDOWN.
7. THE MAXIMUM NUMBER OF CONTAINERS THAT CAN BE LOADED ONTO A TRAILER IS:

   40 THRU 48 FT TRAILERS - 24 CONTAINERS

8. PRIOR TO LOADING THE TRAILER AND DURING PRELOADING INSPECTION REQUIRED BY NAVSEA OP 2165 VOLUME 1 AND REPORTED ON DD FORM 626, THE CHAINS, FITTINGS, AND LOAD BINDERS SHALL BE INSPECTED FOR STRETCH, GOUGING, BENT LINKS, WEAR, AND ANY OTHER NOTICEABLE DEFECTS. THE INSPECTOR SHALL CONFIRM THAT THE CHAINS, FITTINGS, AND LOAD BINDERS HAVE BEEN INSPECTED AND SHALL SO NOTE IN ITEM NO. 22 OF DD FORM 626. ANY DEFICIENCY SHALL BE CAUSE FOR REJECTION OF A CHAIN OR LOAD BINDER.
9. UNLESS OTHERWISE SPECIFIED, DUNNAGE LUMBER, STEEL STRAPPING, STRAP SEALS, AND NAILS SHALL BE AS SPECIFIED IN THE GENERAL TRUCKLOADING DOCUMENT MIL-STD-1320 (NAVY).
10. THE MAXIMUM GROSS WEIGHT OF THE TRACTOR-TRAILER AND THE ALLOWABLE AXLE WEIGHTS ARE THE RESPONSIBILITY OF THE CARRIER. THE CARRIER WILL ADVISE THE SHIPPER OF THESE LIMITATIONS, AND THE SHIPPER SHALL LOAD THE TRAILER IN SUCH A MANNER THAT THE TRACTOR-TRAILER WILL NOT EXCEED THESE LIMITATIONS.
11. AFTER THE BLOCKING, BRACING, AND TIEDOWNS HAVE BEEN INSPECTED, THE CONTAINERS SHALL BE COMPLETELY COVERED WITH FIRE RESISTANT, WATERPROOF TARPAULINS. THE TARPAULINS MAY BE UNDER THE TIEDOWNS AND THE PROTECTOR BOARDS.
12. AFTER THE TARPAULINS ARE IN PLACE, ATTACH THE PROPER EXPLOSIVES PLACARDS TO BOTH SIDES, FRONT, AND BACK OF THE TRAILER, AND ATTACH THE SHIPPING DOCUMENTS TO AN ACCESSIBLE AREA ON THE BACK DECK OF THE TRAILER.
13. FOR GENERAL TRUCKLOADING PROCEDURES REFER TO THE GENERAL TRUCKLOADING DOCUMENT MIL-STD-1320 (NAVY).

UNITIZING PROCEDURE



DETAIL A

1. USING A FORKLIFT TRUCK, STACK TWO CONTAINERS TOGETHER AS SHOWN ABOVE. ORIENT THE TOP CONTAINER OPPOSITE THE BOTTOM CONTAINER (AFT END OF THE TOP CONTAINER OVER THE FORWARD END OF THE BOTTOM CONTAINER).
2. SECURE BOTTOM CONTAINER TO THE TOP CONTAINER WITH FIVE 1-1/4 X .035 STRAPS POSITIONED AS SHOWN IN DETAIL A.
3. TENSION AND DOUBLE NOTCH THE SEALS.

WARNING

TO AVOID TOPPLING, SPECIAL CARE SHOULD BE TAKEN WHEN MOVING STACKS OF CONTAINERS, ESPECIALLY WHEN THE SURFACE IN THE DIRECTION OF THE MOVE IS UNEVEN OR NOT LEVEL. THE UNITIZING OF THE CONTAINERS IS ONLY AUTHORIZED AS PART OF THE TRUCKLOADING PROCEDURE. MOVEMENT OF UNITIZED CONTAINERS SHALL BE LIMITED TO THAT NECESSARY TO LOAD OR UNLOAD THE TRAILER. THE CENTER OF GRAVITY OF CONTAINERS LOADED WITH THE ROCKET MOTORS IS TOWARDS THE AFT END OF THE CONTAINER. WHEN HANDLING THE LOADED CONTAINER(S) WITH A FORKLIFT TRUCK, EXERCISE APPROPRIATE CAUTION.

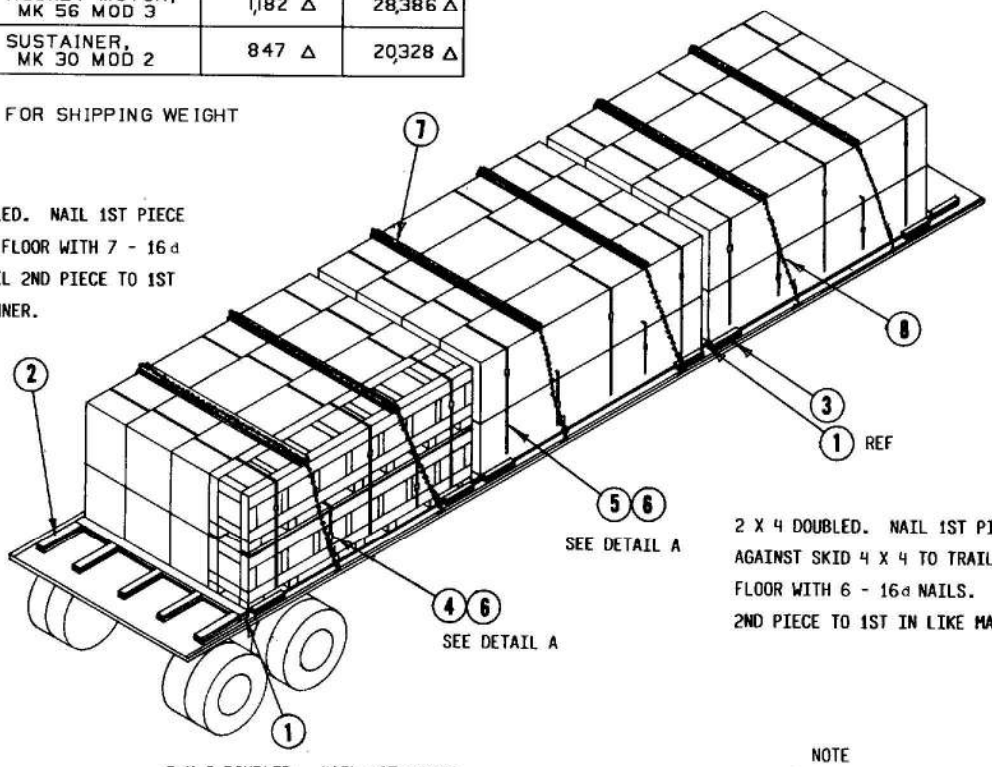
Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:44Z
Check the source to verify that this is the current version before use.

TABLE I

DODIC/NALC	ITEM	GROSS WT. (LBS)	FTL WT. (LBS)
V920	ROCKET MOTOR, MK 104 MOD 1	1,349 Δ	32,376 Δ
V924	ROCKET MOTOR, MK 104 MOD 2	1,349 Δ	32,376 Δ
ZW36	ROCKET MOTOR, MK 56 MOD 0, 1	1,182 Δ	28,386 Δ
V878	ROCKET MOTOR, MK 56 MOD 2	1,182 Δ	28,386 Δ
WW51	ROCKET MOTOR, MK 56 MOD 3	1,182 Δ	28,386 Δ
V284	SUSTAINER, MK 30 MOD 2	847 Δ	20,328 Δ

Δ DO NOT USE FOR SHIPPING WEIGHT

FTL 40 FT FLATBED TRAILER



2 X 6 DOUBLED. NAIL 1ST PIECE TO TRAILER FLOOR WITH 7 - 16d NAILS. NAIL 2ND PIECE TO 1ST IN LIKE MANNER.

2 X 4 DOUBLED. NAIL 1ST PIECE AGAINST SKID 4 X 4 TO TRAILER FLOOR WITH 6 - 16d NAILS. NAIL 2ND PIECE TO 1ST IN LIKE MANNER.

2 X 6 DOUBLED. NAIL 1ST PIECE TO TRAILER FLOOR WITH 1 - 16d NAIL EVERY 6 - 8 INCHES. NAIL 2ND PIECE TO 1ST IN LIKE MANNER.

NOTE
TARPAULIN OVER CONTAINERS OMITTED FOR CLARITY.

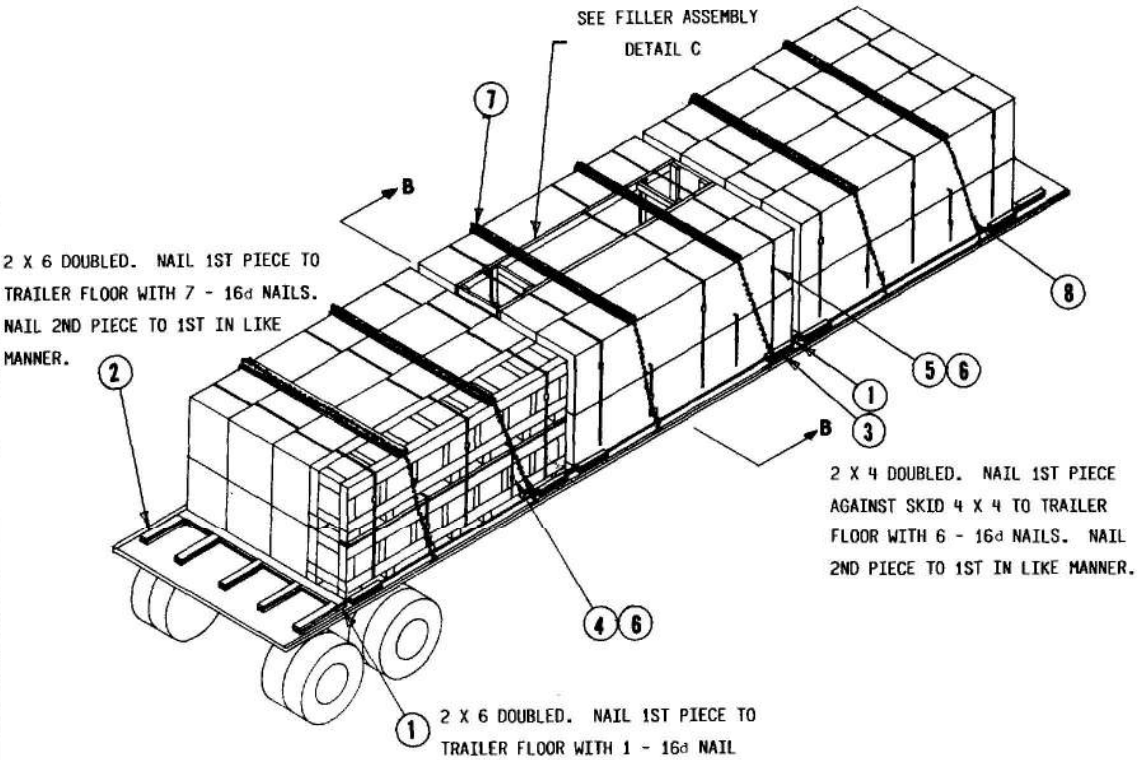
REQ'D	ITEM	DESCRIPTION	MAT'L/DWG	DIMENSIONS
6	8	CHAIN & LOAD BINDER		3/8 OR 5/16
12	7	PROTECTOR BOARD	WOOD.	2 X 6 X 80
60	6	SEAL		1-1/4 X .035
36	5	LONG STRAP	STEEL.	1-1/4 X .035 X 150
24	4	SHORT STRAP	STEEL.	1-1/4 X .035 X 100
24	3	SLEEPER	WOOD.	2 X 4 X 24
20	2	BACKUP CLEAT	WOOD.	2 X 6 X 30
8	1	CROSSMEMBER	WOOD.	2 X 6 X 84

LIST OF MATERIALS

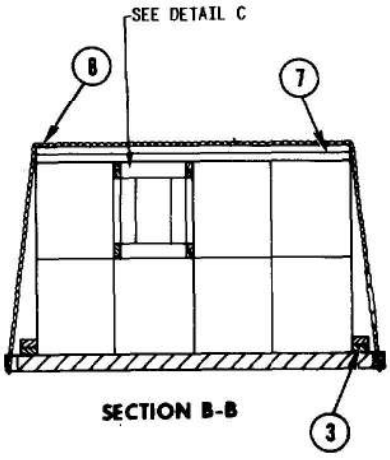
Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:44Z
Check the source to verify that this is the current version before use.

LTL 40 FT FLATBED TRAILERS

NOTE
TARPAULIN OVER
CONTAINERS OMITTED
FOR CLARITY.



LTL 40 FT FLATBED TRAILER



SECTION B-B

REQ'D	ITEM	DESCRIPTION	MAT'L/DWG	DIMENSIONS
8	11	LATERAL SPACER	WOOD.	2 X 4 X 17
8	10	VERTICAL SPACER	WOOD.	2 X 4 X 25
4	9	LONGITUDINAL SPACER	WOOD.	2 X 4 X 110
6	8	CHAIN & LOAD BINDER		3/8 OR 5/16
12	7	PROTECTOR BOARD	WOOD.	2 X 6 X 80
60	6	SEAL		1-1/4 X .035
36	5	LONG STRAP	STEEL.	1-1/4 X .035 X 150
24	4	SHORT STRAP	STEEL.	1-1/4 X .035 X 100
24	3	SLEEPER	WOOD.	2 X 4 X 24
20	2	BACKUP CLEAT	WOOD.	2 X 6 X 30
8	1	CROSSMEMBER	WOOD.	2 X 6 X 86

LIST OF MATERIALS

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:44Z
Check the source to verify that this is the current version before use.

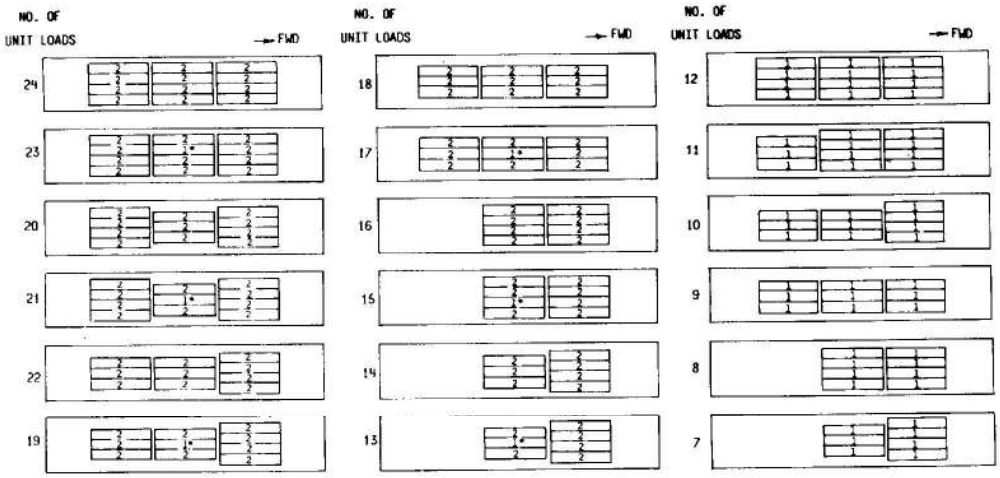
LTL 40 FT FLATBED TRAILERS

1. WHEN REQUIRED TO SHIP LESS-THAN-TRUCKLOAD, SELECT THE CORRECT LOAD PATTERN FROM TABLE II. BLOCK AND BRACE USING PRINCIPLES SHOWN ON PAGES 3, 4 AND 5.
2. FOR STACKS 3 CONTAINERS WIDE, USE CROSSMEMBERS 60 - 66 WIDE, PROTECTOR BOARDS 60 WIDE AND 4 BACKUP CLEATS.
3. FOR STACKS 2 CONTAINERS WIDE, USE CROSSMEMBERS, 40 - 48 WIDE, PROTECTOR BOARDS 40 WIDE AND 3 BACKUP CLEATS.
4. FOR SINGLE STACK, USE CROSSMEMBER 20 - 26 WIDE, PROTECTOR BOARDS 20 WIDE AND 2 BACKUP CLEATS.
5. WHERE 4 CONTAINER WIDE STACKS MEETS 3 CONTAINER WIDE AND 3 WIDE MEETS 2 WIDE, USE PRINCIPLE OF DETAIL D.

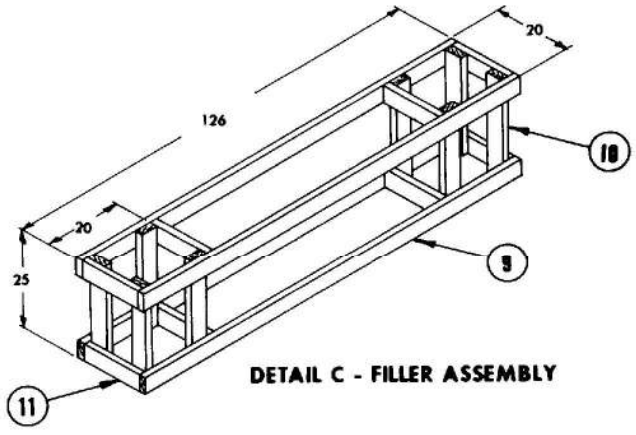
TABLE II

LOAD PATTERN FOR 40 FT FLATBED TRAILERS

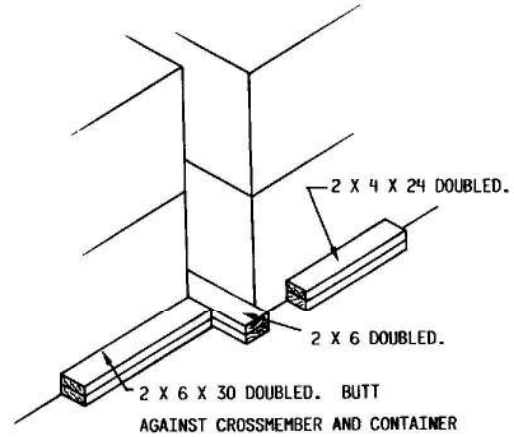
*REQUIRES FILLER ASSEMBLY, DETAIL C, ON TOP AND UNITIZED TO BOTTOM CONTAINER.



6 OR LESS: CONTINUE DECREASING NUMBER OF CONTAINERS AND STACKS OF CONTAINERS AS REQUIRED.



DETAIL C - FILLER ASSEMBLY



DETAIL D

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:44Z
Check the source to verify that this is the current version before use.